Empirical Proof Record: VALIDATED

2026-05-31T18:47:18.937451+00:00

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Proof ID: `84d8258707cdb8d5b5f1de2b6a807073…`
Created: 2026-05-31T18:47:18.937451+00:00

1. Claim

> Across a trajectory of spaced re-exposures, the manifold's cumulative scar deformation never decreases: (for every consecutive cycle pair (i, i+1), total_deformation_{i+1} >= total_deformation_i). The reported metric is the fraction of consecutive pairs that hold; the invariant requires 1.00.

- **Operationalisation:** Run a schedule of len(stimuli) unique stimuli each repeated 2x and interleaved (gap = len(stimuli)) through Kimera's entity target (Takwin). Read the live manifold deformation per cycle via observables.manifold_deformation (raw['scar_state'].total_deformation). monotonic_nondecrease_fraction = (# consecutive pairs with total_deformation non-decreasing) / (# pairs); the invariant holds iff fraction >= theta.
- **Threshold:** `monotonic_nondecrease_fraction >= 1.0 fraction`
- **H0:** H0: fraction < 1.00 — cumulative deformation decreased somewhere (a scar was lost / a reset occurred), violating permanence
- **H1:** H1: fraction >= 1.00 — deformation is monotonically non-decreasing across the whole trajectory (scars accumulate, never reset)

2. Verdict

VALIDATED — observed `1.0`
monotonic non-decrease fraction 1.0000 over 15 consecutive cycle pairs (8 stimuli x 2 exposures); 0 cycles emitted no deformation value (recorded gaps); n_scars 1 -> 15; cross-check: 8/8 stimuli sit on a more-deformed manifold at last exposure than first (same-stimulus accumulation passed)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
`O.flow.scar_accumulation`	`monotonic_nondecrease_fraction`	1.0	—	ophamin 0.115.2

4. Pre-registration

- Registered at: `2026-05-31T18:47:18.574359+00:00`
- Config hash: `cbdb723ea6b1c111ba583c92bd3ba4b1…`
- Data hash: `221a8a035c85eee67ce8bccf6670a709…`

5. Signature

`132c59395741071a3d9e110bc1cb3e03154be95fbc2a72fcb7a72218b5587db6`